Information-optimal illumination for dead-time-limited single-pixel imaging:
certificates, water filling, and the jitter-capped ridge

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Abstract

A photon-counting detector operated at its deterministic information-optimal flux can forfeit more than half of its recorded information the moment its dead time carries a few percent of timing jitter. This paper asks how hard a real, jittery detector can be driven, and with what illumination — one law answers the first half, one certified design principle the second, and a preregistered campaign seals both. The law: for hidden random hold times of coefficient of variation c_v, we derive an exact missing-information identity and the long-window count-information rate $J_\infty(\rho, c_v) = \rho/[(1+\rho)(1+c_v^2\rho^2)]$, whose optimum is finite for every nonzero hold-time variance, solves the cubic $c_v^2\rho^2(1+2\rho) = 1$, and scales as $2^{-1/3}c_v^{-2/3}$ at small jitter; matching this extensive loss to the deterministic terminal-phase loss of a companion study [1] predicts the smooth crossover $\rho^*(\nu, c_v) \sim (6\nu)^{1/3}(1 + 12\nu c_v^2)^{-1/3}$, with zero jitter the critical manifold on which the ridge grows as $\nu^{1/3}$ ($\nu = T/\tau$ dead-time slots per exposure). Exact-law peak predictions registered before the refined Monte-Carlo sweep completed were confirmed out-of-sample at both fresh points (3% and 6%), and a pooled two-run continuous-peak fit measures the jitter exponent at -0.658 (1.3% from $-2/3$). The design principle: hand-designed sparse and matched patterns are atoms of one concave experiment-design program whose per-atom information is the exact renewal kernel, and under deployment safety that program yields a sharply localized doctrine. Information rewards scene-adapted measurement geometry; a uniform $\pm 5\%$ per-pixel dose constraint controls cumulative exposure but does not eliminate that geometry headroom — a development-only constructive probe exhibits a dose-uniform, budget-feasible design with more than twice the local D-information of the balanced family in audited development cells; whether the headroom survives the full deployed conditioning stack — incident budget, dose band, A-risk cap, and spectral floor — is tested by a 480-cell conic near-optimality certificate of the balanced (SCAT32) design; and global operating-point control is orthogonal and survives, one power knob that tracks the jitter-capped ridge. A preregistered simulation campaign (M1) tests the doctrine with a paired fixed-dwell ridge operating-point primary (RIDGE-SCAT32 versus SCAT32-060, a power-for-time comparison), a nine-dwell time-to-quality secondary, the full-stack certificate, five arms, and a mandatory disclosure ledger. The three verdicts are mutually non-rescuing: the ridge operating-point primary passes (median fixed-dwell radiometric gain 1.87 dB at $\nu = 2000$, 19/24 scenes positive, 2.5% bootstrap lower bound 0.12 dB, at a 37× incident-dose multiplier over the balanced comparator); the photon-time-to-quality secondary is a preregistered negative (median speed ratio 0.28, 1/24 scenes above unity), the two operating verdicts jointly instantiating the conjugate photon- and time-budget resource corners; and the full-stack certificate does not certify near-optimality (0/480 cells certified, 299 full-stack-feasible counterexamples exceeding the bar and 181 numerically unresolved), leaving the balanced design's scene-adapted geometry headroom intact under the full conditioning stack. The result is an operating handbook: measure the jitter, let the law set the load, read the certificate as a geometry diagnostic, and tune the global operating point when time is the scarce resource.

1 Introduction

Drive a photon-counting detector to the flux where its count information peaks, and a five-percent timing jitter in its dead time silently forfeits about 55% of that information (Table 1). Single-pixel imaging (SPI) and computational ghost imaging reconstruct a scene from the responses of a single bucket detector to structured illumination [2, 3, 4]; in the photon-starved regime the bucket is a photon-counting device that is blind for a dead time τ after each registered count, and the conventional response is to operate at low flux, keeping the per-slot load $\rho = \lambda\tau$ at the few-percent level and paying in acquisition time. A companion study [1] showed that this convention leaves gains on the table: the exact Fisher information of the integrated nonparalyzable count possesses a principal ridge at the cube-root load $\rho^*(\nu) = (6\nu)^{1/3} - 2/3 + O(\nu^{-1/3})$, and that work mapped when driving a fixed illumination family toward the ridge helps. The trap above is the sharp edge of that map: the deterministic ridge is not where a *real* detector should operate.

Two questions therefore remain open. First, a *deployment* question: how hard can a real detector — whose dead time is a random hold time with some measurable coefficient of variation — actually be driven before the ridge collapses? Second, a *design* question: under a fixed incident-photon and dose budget, which illumination extracts the most information from the dead-time channel, and by how much can a certified optimizer improve on a strong fixed hand design?

One law answers the first question: hidden hold-time randomness adds an *extensive* missing-information term, the long-window optimum becomes finite at the root of an exact cubic, and the deterministic ridge is the zero-jitter critical manifold of a two-parameter crossover (Section 2); its peak predictions, registered before the refined Monte-Carlo sweep existed, were confirmed out-of-sample. One certified design principle answers the second: hand-designed patterns are atoms of a single concave experiment-design program over design measures with the exact renewal kernel; uniform dose alone controls cumulative exposure without eliminating scene-adapted geometry headroom, so the operative restriction is the full deployed conditioning stack, against which the balanced design's near-optimality is tested by a certified conic dual bound over an expanded frozen dictionary, with the global detector operating point as the orthogonal surviving control (Section 3). A preregistered campaign (M1) seals both with a ridge operating-point primary, a time-to-quality secondary, the full-stack certificate, and a mandatory disclosure ledger (Sections 4–5).

We formulate a local approximate-design framework for single-pixel illumination in which each pattern's information is computed from the exact finite-window nonparalyzable renewal count law, optimized under explicit detector-load, dose, per-pixel-exposure, and peak-irradiance constraints and accompanied by generalized-equivalence and exact-rounding certificates. Existing photon-counting optimal experiment design, adaptive or learned SPI, and dead-time optimal-flux studies address neighboring components [5, 6, 7, 8, 9, 10]; we found no prior framework combining these elements for integrated-count single-pixel illumination. The certificate is always stated relative to the frozen dictionary, the local pre-scan estimate, and the declared resource model.

This work contributes: (i) the random-hold count-information law (Section 2): an exact missing-information identity, the long-window rate with its exact jitter-limited cubic optimum and small-jitter coefficient $2^{-1/3}$, and the matched crossover prediction $(6\nu)^{1/3}(1 + 12\nu c_v^2)^{-1/3}$, verified out-of-sample against preregistered predictions with a pooled jitter-exponent measurement of -0.658; (ii) the certified design principle (Section 3): one concave program with the exact renewal kernel, a constructive development-stage localization showing that uniform dose alone leaves scene-adapted geometry headroom, a full-stack conic near-optimality certificate over an expanded gain-coupled dictionary, frozen guards, and the photon-budget versus time-budget conjugacy as its operating doctrine; (iii) a preregistered simulation campaign (M1) whose scenes, atoms, load grid, guards, endpoints, and three mutually non-rescuing verdicts were frozen in an immutable commit before any confirmatory reconstruction existed and are reported as frozen: the ridge operating-point primary passes, the photon-time-to-quality secondary is a preregistered negative, and the full-stack certificate returns confirmatory counterexamples rather than certification; and (iv) an operating handbook for real detectors (Table 7), ending in a bench-testable prediction. All code, frozen preregistration documents, and per-cell evidence bundles are openly released; full derivations, guard machinery, and verification detail are in the supplement.

2 The law: driving a jittery detector

The recovery-time law of the detector determines a scalar count-information kernel; everything else in this paper allocates illumination against that kernel. The companion's deterministic detector [1] is the zero-jitter critical manifold of the theory below. Proofs at manuscript rigor are given in the supplement.

2.1 Random-hold count information

Generalize the detector so that after each registered event it blocks for a hidden *random* hold time $B_i \sim H$, iid, independent of the Poisson arrivals and of λ, with mean τ and coefficient of variation $c_v = \sigma_B/\tau$ (the companion's deterministic model is $c_v = 0$). The observed datum remains the integrated registered count N_T over $T = \nu\tau$; the active time L_T, the state path, and the realized holds are hidden from the count-only observer.

Theorem 1 (missing-information identity) *For $\theta = \log \lambda$, the complete detector-path information is $\mathbb{E}[N_T]$, and the information retained by the scalar count is exactly*

$$I_N(\theta; T) = \mathbb{E}[N_T] - \lambda^2\, \mathbb{E}\big[\mathrm{Var}(L_T \mid N_T)\big] \qquad (1)$$

for every finite T and every iid hidden hold law H independent of λ.

The companion's terminal-phase identity is the special case in which the uncertainty in L_T is confined to the terminal detector phase. The identity requires amendment if the hold law depends on λ or the scene, if realized holds are observed and used, if holds are serially correlated, or if the counter is paralyzable.

Theorem 2 (long-window rate) *Under standard nonlattice renewal local-limit conditions,*

$$\frac{I_N(\log\lambda;T)}{\nu} \longrightarrow J_\infty(\rho,c_v)=\frac{\rho}{(1+\rho)\,(1+c_v^2\rho^2)}, \quad (2)$$

depending on H only through its mean and variance. The missing-information rate converges to $c_v^2\rho^3/[(1+\rho)(1+c_v^2\rho^2)]$: an extensive *loss equal, to first order in $c_v^2\rho^2$, to the heuristic $\lambda^2\sigma_B^2\,\mathbb{E}[N_T]$ — conditioning on the count partially reveals the accumulated hold fluctuation, contributing the factor $(1+c_v^2\rho^2)^{-1}$ — and, unlike the deterministic terminal-phase loss, nonvanishing as $\nu\to\infty$.*

Theorem 3 (exact jitter-limited optimum) *For every fixed $c_v>0$, $J_\infty(\rho,c_v)$ has a unique interior maximum ρ_c solving the cubic*

$$c_v^2\,\rho_c^2\,(1+2\rho_c)=1, \quad (3)$$

with small-jitter expansion

$$\rho_c=2^{-1/3}c_v^{-2/3}-\tfrac{1}{6}+\tfrac{2^{1/3}}{36}\,c_v^{2/3}+O(c_v^{4/3}), \quad (4)$$

whose leading coefficient is $2^{-1/3}=0.79370\ldots$

For exponential holds ($c_v=1$, outside the small-jitter regime) the exact positive root is $\rho_c=0.6573\ldots$; gamma, bounded two-point, and lognormal holds sharing a small c_v share the leading coefficient, with distribution shape entering at the next order.

2.2 The crossover and its universality class

Matching the extensive loss $c_v^2\rho^2$ against the deterministic terminal-boundary loss $\rho^2/(12\nu)$ in the joint regime $\nu\to\infty$, $c_v\to 0$, $c_v^2\nu=O(1)$ gives

$$\rho_*(\nu,c_v)\sim\left(2c_v^2+\frac{1}{6\nu}\right)^{-1/3}=(6\nu)^{1/3}\left(1+12\nu c_v^2\right)^{-1/3}, \quad (5)$$

with scaling variable $x=12\nu c_v^2$ and crossover at $c_v\asymp(12\nu)^{-1/2}$: the deterministic limit $x\to 0$ recovers $(6\nu)^{1/3}$ and the jitter-dominated limit recovers $2^{-1/3}c_v^{-2/3}$. An earlier $\min\{(6\nu)^{1/3},\,c_0\,c_v^{-2/3}\}$ conjecture is only the two-limit envelope of Eq. (5), with an artificial kink, and is superseded. Per the frozen claim discipline, the exact identity and the long-window limit *imply* the jitter exponent $-2/3$; the full finite-window crossover of Eq. (5) is a matched-asymptotic *prediction* until a uniform two-parameter (triangular-array renewal local-limit) proof is complete [supplement: uniform crossover proof, or the prediction label is retained].

The exponent pair $(1/3,-2/3)$ is universal within the regular finite-variance iid hidden-hold class: iid hidden nonnegative holds whose law is independent of λ, with finite nonzero mean and finite variance, forming a regular small-jitter family, under count-only observation and active-start nonparalyzable operation. It is *not* claimed for heavy-tailed (infinite-variance) holds, correlated or drifting holds, holds observed by the electronics, paralyzable detectors, event-time observations, or λ-dependent recovery. The precise critical statement is: zero hold-time variance is the critical manifold on which the optimal load continues to grow with window length; any fixed nonzero hidden hold variance produces a finite long-window optimum. Adjacent renewal-CLT and random-dead-time literature establishes the mean/variance ingredients and task-specific finite-rate optima [11, 12, 13, 14, 10]; we found no source stating the count-only exponent pair or the extensive/terminal crossover.

Table 1: Scalar count information $J(\rho,\nu=2000,c_v)$ for lognormal hold times: three-stage peak measurement against the exact-law predictions of Theorem 3/Eq. (5), registered *before* the data (`docs/R14_PREREGISTERED_PREDICTIONS.md`, commit `8a627bb`; verdicts at `1d8d7aa`, `4d8310d`, `bd731d5`). Grid: `jitter_sfi_v2`, 100,000 frames, 25-point log grid (spacing 1.157). Zoom: continuous quadratic-interpolated peaks, local run, 150,000 frames. Replication: independent Colab run, 400,000 frames. Both out-of-sample (o.o.s.) grid rows hit ($c_v=0.02$: 3%; $c_v=0.2$: 6%); the pooled eight-measurement log-log fit gives slope -0.658 vs. $-2/3$ (1.3%), with the constant consistent with $2^{-1/3}=0.794$ within 5–10% (the earlier coarse-grid fit 0.72 was a finite-grid artifact) and a reproducible mild $\approx -8\%$ low bias at the end points (open higher-order note). Hardware warning: at the deterministic ridge ($\rho=22.3$), 5% jitter leaves $J=0.43$ vs. 0.95 — about 55% of the scalar count information forfeited. J at peak from the grid run.

c_v	peak ρ (grid / zoom / replic.)	exact law	J at peak
0	22.297 / — / —	22.25 (ridge)	0.9452
0.02	10.739 / 9.645 / 9.610	≈ 10.4 (o.o.s.)	0.8731
0.05	5.173 / 5.154 / 5.700	5.69	0.7848
0.1	3.862 / 3.752 / 3.399	3.50	0.6989
0.2	2.153 / 2.106 / 2.051	≈ 2.30 (o.o.s.)	0.5882
0.3	1.607 / — / —	1.63	0.5042

2.3 Preregistered verification

The law was tested the way the campaigns are run: predictions first, data second. The exact-law column of

Table 1, including two out-of-sample rows, was registered before any refined measurement existed (commit `8a627bb`). The grid sweep then confirmed the law at both registered fresh points — $c_v = 0.02$: predicted ≈ 10.4, measured 10.739 (3%); $c_v = 0.2$: predicted ≈ 2.30, measured 2.153 (6%) — with every in-sample peak within one grid notch and the $\nu = 200$ column consistent including the crossover correction of Eq. (5). A continuous-peak zoom and an independent replication (Colab, 400,000 frames) then measured the exponent itself: the pooled eight-measurement log-log fit gives slope -0.658, within 1.3% of the predicted $-2/3$, with the constant consistent with $2^{-1/3}$ within 5–10% and a reproducible mild $\approx -8\%$ end-point low bias left as an open higher-order note (candidate: a lognormal third-moment correction). One honest beat from the record: a mid-sweep effective-multiplier hypothesis ($k = 4/3$) was preregistered and refuted by the very next measured point (commits `5e51088`, `1a32758`). Run specifications, both-run peak tables, and the frozen three-family exponent battery (tests E1–E5) that remains a preregistered extension are given in the supplement.

3 The certified design principle

The design principle is a doctrine in four movements. Information rewards scene-adapted measurement geometry: the dose-relaxed optimum and a constructive dose-only probe expose substantial local D-information headroom (Sections 3.1 and 3.3). Uniform dose is not enough to eliminate that headroom: a balanced cumulative exposure can coexist with scene-adapted measurement directions (Section 3.3). Conditioning safeguards define the deployable class: the A-risk and spectral anchors test whether the geometry headroom survives reconstruction-trust requirements, and the full-stack certificate quantifies what remains (Section 3.4). Global operating-point control is orthogonal and survives: ridge tracking changes one global source multiplier and is evaluated by the empirical power-for-time endpoints (Sections 3.5 and 3.7).

3.1 One concave program over design measures

Freeze the coarse pre-scan reconstruction at $\hat{x}$ and parameterize the locally estimable image perturbation as $x(\theta) = \hat{x} + B\theta$, with $B \in \mathbb{R}^{n\times r}$ spanning a declared r-dimensional estimable subspace. For a nonnegative illumination atom a the arrival rate is $\lambda(a) = \Phi\, a^\top \hat{x} + d$ and the load is $\rho(a) = \tau\lambda(a)$. With $J_{\mathrm{exact}}(\rho, \nu)$ the exact per-slot information of the integrated renewal count for $\log\lambda$ (generalized to the random-hold kernel of Theorem 2 when $c_v > 0$) and score direction $q(a) = \Phi\, B^\top a/\lambda(a)$, the rank-one local information atom is the chain-rule form

$$
\begin{aligned}
H(a;\hat{x}) &= \nu\, J_{\mathrm{exact}}(\rho(a), \nu)\, q(a) q(a)^\top \\
&= \nu\, J_{\mathrm{exact}}(\rho(a), \nu)\, \frac{\Phi^2}{\lambda(a)^2}\, B^\top a a^\top B.
\end{aligned}
\tag{6}
$$

A raw sensitivity $g \propto \Phi a$ is incomplete unless the factor $\sqrt{\nu J_{\mathrm{exact}}}/\lambda$ is absorbed into it. For a probability measure ξ over the compact feasible atom set $\mathcal{A}$, define $V(\xi) = V_0 + \int_{\mathcal{A}} H(a)\, d\xi(a)$ with $V_0 \succeq 0$ the frozen pre-scan information, and maximize the local D-criterion

$$
\max_{\xi}\ \log\det V(\xi) \quad \text{s.t. linear resource constraints,} \tag{7}
$$

subject to average load, cumulative dose, family allocation, and peak constraints. The objective is concave in ξ; it is a *local information surrogate* and is not claimed to optimize reconstruction PSNR or any image-domain risk. The estimable subspace is frozen at $r = 200$ (the top eigenbasis of the fixed comparator's information matrix, fixed before the solve, with a fixed numerical ridge; conventions in the supplement). If every atom obeys a pointwise equal-load constraint the scalar kernel factors out and the geometry reduces to ordinary linear D-optimality [15]; genuine dead-time-aware design therefore requires controlled load variation. Unconstrained, this geometry rewards scene-adapted concentration of measurement effort on the brightest, highest-value directions — the incentive whose fate under deployment safety the rest of this section traces.

3.2 Safety guards and the retirement of the adaptive arm

The scene-adapted incentive collides with the deployed safety guards in the audited development evidence. The earlier variable-load adaptive arm (OED-DT) is retired as a proposed or gated imaging method: its speed gate was withdrawn before the campaign freeze as infeasible, not recorded as a confirmatory failure. Within the frozen target-load-normalized atom family, under the declared guard stack — incident budget, ±5% per-pixel dose, A-risk cap, and spectral floor — no nonzero adaptive mixture is admissible in the audited development cell: the descending exact mixture scan terminates at zero adaptive rows (`ADAPTIVE_COLLAPSE_UNDER_GUARDS`), every nonzero mixture violating at least one of the dose, A-risk, or spectral guards. This is development evidence about one parameterization under the joint guard stack, explicitly not a confirmatory theorem; the target-load-normalized atomization fixes each pattern's amplitude to hit a predicted load, which on dim supports demands large physical amplitude and anti-aligns the dose vector with brightness. Two consequences are drawn, and they are separated deliberately. First, the constructive development probe of Section 3.3 shows that the dose constraint *alone* does not produce the collapse: dose-uniform, budget-feasible geometry headroom exists. The operative restriction is therefore the conditioning stack, and the near-optimality certificate of Section 3.4 targets exactly that full stack over an *expanded* gain-coupled class. Second, the descending scan survives only as a path-specific descriptive diagnostic (`PATH_FEASIBLE_ALPHA`, relative to the frozen row ordering, reported in the methods-development supplement), never an estimator, fallback, or certificate. The dose-relaxed and dose-constrained continuous OED solutions are retained as no-endpoint design diagnostics that illustrate the incentive and the safety frontier (Figure 1); neither is the proposed imaging method.

3.3 Development evidence: the dose-only constructive probe

Development-only constructive analysis; not a confirmatory endpoint or population estimate. A development-only constructive probe showed that the $\pm 5\%$ per-pixel dose constraint alone did not eliminate scene-adapted geometry selection: a feasible low-load geometry design achieved a substantially larger local determinant than SCAT32 in the audited cells. This was a lower-bound construction, not an estimate of the dose-only optimum. Concretely, on the glyph development scene (`m1_dev_glyph`, measurement seed 0), at the two audited cells $(\nu, b) = (2000, 0.60)$ and $(200, 0.05)$, the probe produced probability designs over the frozen D_{cert} satisfying the incident budget and the $\pm 5\%$ dose band with log-determinant improvements over deployed SCAT32 of $\Delta \log\det V/r = 0.776$ and 0.903, certifying local D-efficiency ratios of at least $e^{0.776} \approx 2.17$ and $e^{0.903} \approx 2.47$, respectively. The construction is existence-only, and its composition must not be over-read: the probe initializes positive mass only on the lowest-load D_{load} column, and its multiplicative update is support-preserving (an exactly zero column can never activate), so the reported designs — all mass on D_{load}, uniform low support loads, roughly half the photon budget spent — establish neither that the dose-only optimum avoids gain atoms nor that it prefers low loads or a partial budget. The dose-only class is retained only as a no-gate development analysis; no confirmatory dose-only verdict exists. A frozen six-image replication (all six development images, seed 0, the same two anchor cells, every result reported without selection) is given in Table 2: the per-dimension log-determinant gap ranges from 0.496 to 0.921 across all six families and both anchors, certifying dose-feasible local D-efficiency lower bounds of 1.64× to 2.51×, with every construction budget- and dose-feasible at every recorded point. The dose-only construction shows that uniform exposure still permits scene-adapted geometry selection. Optimizing and validating that geometry under relaxed or task-specific conditioning constraints is left to a separate follow-up; no such arm or endpoint is added to the present campaign.

Table 2: Development-only constructive dose-only probe, frozen six-image replication (`results/round63_m1/R18_GAP_PROBE_REPLICATION.md`; DEV images, seed 0; *not a confirmatory endpoint or population estimate*). Every result is reported without selection. Δ/r is the per-dimension log-determinant gap of the probe design over deployed SCAT32; $\mathrm{Eff}_D^{\downarrow} = e^{\Delta/r}$ is the certified dose-feasible D-efficiency *lower* bound. Constant across all cells (support-preserving initialization caveat, main text): D_{gain} mass 0.000, per-pixel dose deviation 0.038, budget used equal to the low-load quantile (0.300 at $(\nu, b) = (2000, 0.60)$, 0.025 at $(200, 0.05)$); all cells budget- and dose-feasible.

scene	ν	b	SCAT32	probe	Δ/r	$\mathrm{Eff}_D^{\downarrow}$
glyph	2000	0.60	3646.425	3801.577	0.776	2.17×
glyph	200	0.05	2788.834	2969.346	0.903	2.47×
chirp	2000	0.60	3646.994	3819.047	0.860	2.36×
chirp	200	0.05	2788.149	2887.359	0.496	1.64×
maze	2000	0.60	3633.386	3750.598	0.586	1.80×
maze	200	0.05	2790.259	2892.347	0.510	1.67×
spokes	2000	0.60	3640.443	3820.941	0.903	2.47×
spokes	200	0.05	2800.765	2958.083	0.787	2.20×
contour	2000	0.60	3648.639	3829.357	0.904	2.47×
contour	200	0.05	2772.045	2891.034	0.595	1.81×
microtexture	2000	0.60	3667.825	3851.978	0.921	2.51×
microtexture	200	0.05	2810.246	2953.895	0.718	2.05×

3.4 Conditioning safeguards define the deployable class: the full-stack certificate

The deployable class is defined by the conditioning safeguards, and the balanced design's near-optimality within it is certified, not trusted. By concavity, optimality is equivalent to a bounded reduced sensitivity (the constrained Kiefer–Wolfowitz / general equivalence condition for nonlinear models [15, 16]). The frozen certificate wording is exact. The confirmatory certificate therefore targeted the full deployed stack: the frozen dictionary and local pre-scan linearization, incident budget, $\pm 5\%$ per-pixel dose band, A-risk cap, and spectral floor. Global ridge tracking was evaluated separately as the time-limited, power-available control. Explicitly, the certified class $\mathcal{C}_{\text{stack}}$ contains every probability design over the expanded dictionary $D_{\text{cert}} = D_{\text{load}} \cup D_{\text{gain}}$ — the frozen target-load-normalized atoms plus a certificate-only gain-coupled family in which each base geometry is scaled by $\gamma \in \{0.2, 0.5, 1, 2, 5\}$ with its predicted load allowed to *emerge* rather than be rescaled to a target, so fixed-flux self-water-filling and bounded per-pattern gain control lie inside the comparison class — that satisfies the incident budget, the ±5% per-pixel dose band, the A-risk cap $\mathrm{tr}\{V(\xi)^{-1}\} \le 1.05\,\mathrm{tr}(V_{\mathrm{fix}}^{-1})$, and the spectral floor $V(\xi) \succeq 0.5\,V_{\mathrm{fix}}$, with V_{fix} the load-matched deployed SCAT32 information matrix. The A-risk and spectral conditions are *global design constraints* of the class, represented exactly as linear matrix inequalities and priced in the dual by positive-semidefinite multipliers; they are never approximated by dropping individual atoms. The certificate quantity is the support-function upper bound G_{stack} on the log-determinant gap between the deployed design and the continuous optimum of $\mathcal{C}_{\mathrm{stack}}$, computed by a certified cutting-plane outer approximation (spectral eigenvector cuts, convex A-risk tangent cuts, budget/dose prices, column generation, and an exact full-dictionary scan; a finite cut set is an outer relaxation, so a small bound remains conservative); the frozen bar is $G_{\mathrm{stack}}/r \le 10^{-2}$, certifying local D-efficiency of at least $\exp(-0.01) = 0.99005$ (full construction, primal discriminator, and numerical protocol in the supplement). The certificate is comparator- and subspace-relative by construction: it preserves the predeclared conditioning of SCAT32 on the directions SCAT32 resolves, and it is not a global optimum over all images, subspaces, risk tradeoffs, or physical patterns. Continuous weights are rounded to an exact 972-row design by efficient rounding and deterministic exchange [17, 18, 19], admissible only at exact D-efficiency ≥ 0.95. Every remaining guard is a frozen constraint, not a tunable proxy (Table 3); the full guard mathematics, the expanded dictionary listing, and the budget accounting are in the supplement.

Table 3: The frozen design machinery at a glance (R10/R11/R13/R17/R18; full statements in the supplement).

Element	Frozen rule
Dictionary	all cyclic translates of the 13 R10 support families (scattered $k = 16/32$, solid 4×4, Lblob, bars, compact-32, ring); weight powers $p \in \{0, 1\}$; the certificate scan adds gain-coupled copies $a_{g,\gamma} = \gamma a_g^{\mathrm{base}}$, $\gamma \in \{0.2, 0.5, 1, 2, 5\}$, emergent load ($D_{\mathrm{cert}} = D_{\mathrm{load}} \cup D_{\mathrm{gain}}$)
Budget	incident-signal inequality $\int \rho_s\, d\xi \le 0.60$; detected counts dual-reported, never the sole budget
Peak	shape: $w_j \in [\frac{1}{4}, 4]$, $k \ge 16$; physical: $\max a_{ij} \le 1536$
Dose	per-pixel mean exposure within ±5% of image mean, re-verified post-rounding
Floors	spectral $V(\xi) \succeq 0.5\,V_{\mathrm{FIXED}^*}$; A-risk $\mathrm{tr}(V^{-1}) \le 1.05\,\mathrm{tr}(V_{\mathrm{FIXED}^*}^{-1})$ — global design constraints of the certified class $\mathcal{C}_{\mathrm{stack}}$, LMI-priced in the certificate dual
Kernel trust	exact-PMF certification; $J_{\mathrm{mean}}/J_{\mathrm{exact}} \ge 0.90$; RQL log-load bias ≤ 0.05; $p_{\mathrm{ceil}} \le 0.05$ per atom, ≤ 0.01 design-weighted
Certificate	$G_{\mathrm{stack}}/r \le 10^{-2}$ (full-stack conic dual, expanded D_{cert} scan; per-cell status `CERTIFIED` / `COUNTEREXAMPLE` / `NUMERICAL_UNRESOLVED`; local D-eff. ≥ 0.99005)
Rounding	exact 972-row design, $\mathrm{Eff}_D \ge 0.95$

3.5 The conjugacy doctrine: photon budget versus time budget

The doctrine has two conjugate resource corners of the same dead-time information map. In the *photon-budgeted* corner the full-stack certificate holds the incident-signal resource as a hard *inequality*,

$$\int \rho_s(a)\, d\xi(a) \le 0.60, \tag{8}$$

so a design measure may push a subset of high-value atoms into the ridge zone only if compensated by low-load or omitted atoms (R11 §4); detected counts are dual-reported but are never the sole fairness budget, because $\mathbb{E}[N \mid \rho] \simeq \nu\rho/(1+\rho)$ has vanishing marginal cost at high load. The *time-budgeted* corner is the no-gate ridge-tracking policy RIDGE-SCAT32: it fixes the balanced support and applies one global multiplier per dwell, calibrated on the pre-scan estimate to mean load $\rho_R(\nu)$, followed by a single global safety clip, spending a deliberately larger incident budget for time rather than photons (R11 §2). The exact production ridge is $\rho_R(\nu) = \min \arg\max_{\rho \in \mathcal{R}_{\mathrm{adm}}(\nu)} J_{\mathrm{exact}}(\rho, \nu)$ over the admissible set $\mathcal{R}_{\mathrm{adm}}(\nu) = \{0 < \rho \le 24 :\ p_{\mathrm{ceil}} \le 0.01,\ J_{\mathrm{mean}}/J_{\mathrm{exact}} \ge 0.90\}$, using the exact numerical ridge rather than the asymptotic cube-root form (R11 §1). Under the incident-photon budget the relevant scalar efficiency is $J_{\mathrm{exact}}(\rho, \nu)/\rho$, which decreases across the admitted palette; a photon-budgeted optimizer therefore rationally spends on lower loads and *avoids* the ridge, while the ridge belongs to the time-limited, power-available corner. This is a resource-regime statement, not a universal prohibition on ridge-zone allocation: sufficiently valuable directions may still receive larger loads in other cells (R13 §8(f)).

3.6 One program behind every optimum

Every true interior optimum of the declared design problem — over load, support scale, and intensity weighting — satisfies one generalized reduced-sensitivity/KKT system of the master program, whose load coordinate is the information-water-filling equation and whose scale and weighting coordinates price conditioning through the V^{-1} geometry (full system in the supplement); this also explains why raising proxy contrast from $p = 1$ to $p = 2$ can lower the true objective. One caveat is kept deliberately: the master design problem places load, support scale, and intensity weighting in one constrained concave measure optimization, and its support atoms satisfy a common reduced-sensitivity condition; but the empirically observed occupancy ($k \approx 32$) and weighting ($p \approx 1$) optima *motivate* these coordinates and are not themselves claimed as continuous KKT theorems — they were selected from discrete grids and judged by image endpoints, not by the master $\log\det$ objective.

3.7 Why one global knob suffices for RIDGE-SCAT32

Fixed flux performs implicit information water filling when bucket brightness and local directional value satisfy the marginal-information alignment condition $\ell_i\, J'(\kappa u_i) = \beta$ across active directions ($\ell_i = q_i^\top V^{-1} q_i$ the directional value, $u_i = a_i^\top x$ the bucket brightness); its adequacy is measured by a projected KKT residual — not C_u alone — and outside that condition, variable-load OED can improve without any universal bound on the fixed-flux approximation ratio. On nonnegative sparse-support scenes the source-induced load ordering naturally tracks directional value, which is why RIDGE-SCAT32 needs only a single global multiplier, and why the uniform per-pattern servo — which loses by up to ~ 7 dB in the companion's dev evidence [1] — fails by erasing exactly that alignment. With random holds, J' vanishes at the jitter cap and is negative beyond it, so fixed flux can overdrive the brightest directions and explicit gain control becomes necessary. The full proposition, its residual near-optimality bound, and its failure modes are in the supplement. Naive per-pattern load equalization was thus refuted (R11) for destroying this brightness–information alignment, while the development evidence of Section 3.2 shows the opposite extreme — adaptive concentration — colliding with the deployed conditioning stack; the balanced global-flux policy is the robust middle of the tested extremes, with dose-feasible geometry headroom outside the conditioning stack left to a separate follow-up (Section 3.3).

4 Preregistered campaign (M1)

M1 is a separate follow-up campaign with its own seed namespace, manifests, and expected-cell ledger; it does not reopen the companion's frozen studies. It freezes on the R13 audit plus one immutable commit (tag `m1-freeze`); after that commit no scene, atom family, load grid, guard, endpoint, or gate may change, and if the primary fails there is no third redesign.

Geometry and constants. $n = 32^2 = 1024$; $M_{\text{total}} = 1024 = M_{\text{pre}}(52) + M_{\text{main}}(972)$ for every arm; nonparalyzable active start, $\tau = 50$ ns, $d = 0$, $\sigma_b = 0$; $\bar\rho \in \{0.05, 0.60\}$ for the budget-anchored arms; $\nu \in \{5, 10, 20, 50, 100, 200, 500, 1000, 2000\}$; five measurement seeds per (image, condition). Optical time $T_{\text{opt}} = M_{\text{total}}\, \nu\, \tau$; design-computation wall time is reported separately and never folded into T_{opt}.

Scenes. Twenty-four fresh confirmatory instances (six generator families $\times$ four instances) with seeds $s_{f,r} = 633000 + 100f + r$, plus six development-only instances $632900 + f$. The six committed generators (glyph, chirp, spokes, maze, contour, microtexture) are reused verbatim with their actual names — "USAF" is not used unless a USAF target is generated, and spokes is described as a radial resolution target. No acceptance filter of any kind is applied; generators, images, and ROIs receive SHA256 manifests.

Common pre-scan. Every arm receives and pays for the same 52-pattern scene-independent balanced pre-scan at its own $(\bar\rho, \nu)$: 32 paired 4×4-block rows (support 32, amplitude 32), 16 block rows (amplitude 16), and 4 quadrant rows (amplitude 4), so every pixel receives cumulative amplitude $32 + 16 + 4 = 52$ and the row-mean pattern is exactly the all-ones vector; the mean pre-scan load then equals the declared global load for any sum-normalized scene, with no $\hat{x}$-dependent row scaling (R13 §2). All 52 counts enter every arm's final reconstruction; RIDGE-SCAT32 additionally uses the pre-scan estimate to calibrate its single global source multiplier, with no per-pattern servo. Every arm runs the same 972 SCAT32 main patterns and totals exactly 1024 physical exposures.

Arms and the deployed design. The deployed spatial design is the balanced exact 972-row SCAT32 multiset, identical across every operating-point comparison; the common 52-row pre-scan is charged to each arm. Table 4 lists the five arms. Three share this design and differ only in source power: SCAT32-SAFE at global mean load 0.05, SCAT32-060 at 0.60, and RIDGE-SCAT32, which applies one global multiplier per dwell (calibrated on the pre-scan to $\rho_R(\nu)$, then the frozen global clip). Two frozen fixed baselines, SCAT16 and LBLOB16 at load 0.60, are retained as descriptive context columns (no gate), so the cross-arm occupancy-ladder comparison keeps its reference columns. The $k = 32$ occupancy rung of the deployed design is fixed by the frozen development rule (median fixed-dwell radiometric PSNR at $(\bar\rho, \nu) = (0.60, 2000)$, averaged over development seeds), which selected SCAT32 over the clustered LBLOB16 and a matched baseline with development scores 17.498, 11.908, and 9.181 dB respectively; an earlier information-score-based selection was invalidated by external audit (R15) for departing from the frozen rule and was re-run under the original R10 prescription. The rung is consistent with the compan-

Table 4: The five M1 arms over the common balanced SCAT32 design. The paired RIDGE-SCAT32 vs. SCAT32-060 comparison carries the primary gate and RIDGE-SCAT32 vs. SCAT32-SAFE the speed gate; the full-stack certificate (`FULL_STACK_CERT_PASS`) applies to the deployed design itself. SCAT16 and LBLOB16 are descriptive context.

Arm	Role	Gate
SCAT32-SAFE	balanced design at load 0.05 (safe reference; speed baseline)	speed base
SCAT32-060	balanced design at load 0.60 (primary comparator)	primary comp.
RIDGE-SCAT32	one global multiplier per dwell to $\rho_R(\nu)$ + global clip (ridge policy)	primary, speed
SCAT16	frozen scattered $k = 16$ at load 0.60 (companion bridge)	none
LBLOB16	committed clustered support at load 0.60 (spectral-floor reference)	none

ion's Study 2 fixed-dwell optimum [1]. Reconstruction (RQL with the frozen analytic λ_{TV} rule and pooled initialization) is identical across arms.

Endpoints and gates. Three preregistered, mutually non-rescuing verdicts govern the campaign. The *primary* is a paired fixed-dwell ridge operating-point test (`RIDGE_OPERATING_PASS`). At terminal dwell $\nu = 2000$, for each confirmatory image j define $\Delta Q_j^{\mathrm{ridge}} = \overline{\mathrm{PSNR}}_{\mathrm{rad}}(\text{RIDGE-SCAT32}, \nu{=}2000, j) - \overline{\mathrm{PSNR}}_{\mathrm{rad}}(\text{SCAT32-060}, \nu{=}2000, j)$, the mean over the five frozen seeds, with identical pre-scan, identical 972-row SCAT32 multiset, paired pattern/noise seeds, the same reconstruction path, and the same optical dwell — only the single global main-acquisition multiplier differs. PASS iff $\mathrm{median}_j\, \Delta Q_j^{\mathrm{ridge}} \geq 1.0$ dB, the 2.5% bootstrap lower bound on the median exceeds 0, and $\#\{j : \Delta Q_j^{\mathrm{ridge}} > 0\} \geq 18/24$, using the existing 10,000-replicate nested family-stratified bootstrap; any numerical or guard failure stays in the intention-to-treat cohort and counts as nonpositive. This is a fixed-dwell, power-for-time operating-point test. It is not an equal-photon-budget or photon-efficiency comparison. The *secondary* is the nine-dwell Q_{90} time-to-quality test (`RIDGE_SPEED_PASS`): SCAT32-SAFE at load 0.05 against the dwell-dependent RIDGE-SCAT32 policy, using the frozen PAVA/censoring machinery (the seed-mean radiometric PSNR, equal-weight PAVA isotonic curves, 0.50 dB safe-range informativeness floor, and censoring taxonomy transfer verbatim), with median $S_{\mathrm{gate}} \geq 3$, bootstrap lower bound > 1, and $\geq 18/24$ images with $S_{\mathrm{gate}} > 1$. The *confirmatory structural secondary* is the full-stack near-optimality certificate (`FULL_STACK_CERT_PASS`, Section 3.4). Every one of the $24 \times 5 \times 2 \times 2 = 480$ certificate cells ($\nu \in \{200, 2000\}$, $b \in \{0.05, 0.60\}$) receives exactly one terminal status — `CERTIFIED` (deployed design and class primal-feasible, all residual checks passed, $G_{\mathrm{stack}}/r \leq 10^{-2}$), `COUNTEREXAMPLE` (a concrete full-stack-feasible design with actual log-determinant improvement per dimension $> 10^{-2}$, found by the frozen support-expanding primal screen), or `NUMERICAL_UNRESOLVED` (neither conclusion within the frozen numerical protocol) — under a frozen per-cell wall cap of 420 s (a 60 s primal screen, a 180 s cutting-plane solve, and one deterministic rescaled retry; no third attempt, no threshold change, no imputation, no cell deletion). PASS iff all 480 cells are `CERTIFIED`; `NUMERICAL_UNRESOLVED` means not certified, not scientifically refuted. No verdict can rescue another.

Disclosure ledger. Every cell discloses incident and detected photons (predicted and realized), the incident-dose and detected-count ratios, source/peak-irradiance ratio, the achieved mean load and its quantiles ρ_5, ρ_{50}, ρ_{95}, $\rho_{\max}$, the physical peak, predicted and realized ceiling fractions, any `RIDGE_GUARD_CLIPPED` flag, and mean J_{exact}; certificate cells add the terminal status, the full-stack dual bound G_{stack}, the exact D-efficiency, solver status for the primary solve and the single retry, wall seconds, coefficient dynamic range, dictionary-scan and A-risk/spectral cut counts, the full-dictionary scan residual, the primal and dual gaps per r, the minimum generalized eigenvalue, the A-risk ratio, and the `MU_CAP_ACTIVE` disclosure. The full nine-dwell kernel/ridge certification table and the SHA256 of $D_{\mathrm{cert}} = D_{\mathrm{load}} \cup D_{\mathrm{gain}}$ are frozen before any confirmatory scene is opened.

5 Results

5.1 Primary: ridge operating-point gain

The primary verdict is `RIDGE_OPERATING_PASS` = **PASS**. At $\nu = 2000$ the median paired radiometric gain of RIDGE-SCAT32 over SCAT32-060 is $\mathrm{median}_j\, \Delta Q_j^{\mathrm{ridge}} = 1.87$ dB (frozen bar ≥ 1.0 dB); the 2.5% nested family-stratified bootstrap lower bound on the median is 0.12 dB (bar > 0); and 19/24 scenes are positive (bar $\geq 18/24$). All three conditions pass. The gain is bought with incident power, not photons: the ridge policy reaches its target mean load $\bar{\rho} = \rho_R(2000) = 22.25$ (the global safety clip does not bind), a 37× incident-dose ratio and a 2.6× detected-count ratio over the balanced SCAT32-060 comparator, with

Table 5: Per-family primary result at $\nu = 2000$: median paired ridge gain $\Delta Q^{\mathrm{ridge}}$ and the count of positive scenes (four instances per family). The pooled median is 1.87 dB and 19/24 scenes are positive (RIDGE_OPERATING_PASS). Source: results/round63_m1/M1_VERDICTS.json and the imaging shards.

Family	median $\Delta Q^{\mathrm{ridge}}$ (dB)	scenes > 0
maze	+6.37	4/4
glyph	+3.58	4/4
microtexture	+1.49	4/4
chirp	+1.40	4/4
spokes	+0.71	2/4
contour	−4.18	1/4

no count-ceiling saturation (predicted $p_{\mathrm{ceil}} < 10^{-6}$; realized dark-count and timing-error fractions 0) and no RIDGE_GUARD_CLIPPED event. The gain is strongly scene-dependent (Table 5), from +6.37 dB on maze to −4.18 dB on contour, the sole negative-median family (Section 6). This is a fixed-dwell, power-for-time operating-point test. It is not an equal-photon-budget or photon-efficiency comparison.

5.2 Secondary: ridge time-to-quality

The secondary verdict is RIDGE_SPEED_PASS = **FAIL**, a preregistered negative reported plainly. Against the SCAT32-SAFE reference at load 0.05, the dwell-dependent RIDGE-SCAT32 policy has median photon-time-to-quality ratio $S_{\mathrm{gate}} = 0.276$ (frozen bar ≥ 3), 2.5% bootstrap lower bound 0.172 (bar > 1), and only 1/24 scenes with $S_{\mathrm{gate}} > 1$ (bar $\geq 18/24$); none of the three conditions passes. The direction of the failure is mechanistic, not anomalous. The Q_{90} secondary is measured in optical (photon) time $\nu\rho$, where the ridge policy's large incident-power multiplier — a 37× incident-dose ratio over the balanced comparator at $\nu = 2000$, and larger still over the 0.05 safe reference — dominates, so the fixed-dwell quality gains of the primary do not survive photon-time normalization. The two operating verdicts therefore instantiate the conjugate resource corners of Section 3.5 empirically: when power is available and time is scarce, the ridge wins fixed-dwell quality (primary PASS); when the photon budget is scarce, the low-load balanced design wins time-to-quality (secondary FAIL). Neither rescues the other.

5.3 Confirmatory certificate: near-optimality over the full deployed stack

The confirmatory structural secondary does not certify: FULL_STACK_CERT_PASS = **FAIL**. Of the 480 frozen certificate cells (24 confirmatory scenes ×5 seeds $\times \nu \in \{200, 2000\} \times b \in \{0.05, 0.60\}$), 0 are CERTIFIED, 299 are COUNTEREXAMPLE, and 181 are NUMERICAL_UNRESOLVED; the fraction certified at $G_{\mathrm{stack}}/r \leq 10^{-2}$ is 0. The gate requires 480/480 CERTIFIED, so it fails. The status split is stable across the four anchors ($(\nu, b) = (200, 0.05)$: 77/43; $(200, 0.60)$: 67/53; $(2000, 0.05)$: 84/36; $(2000, 0.60)$: 71/49 counterexample/unresolved). Every cell is deployed-design feasible for budget, dose (±5%; realized deviation 0.040), A-risk, and spectral constraints, with no active finite dual cap (MU_CAP_ACTIVE false in all 480), and the protocol runs well inside its budget (median wall time 64.6 s, maximum 67.3 s, against the 420 s per-cell cap; no cell reaches it). The 299 counterexamples are concrete full-stack-feasible designs whose actual per-dimension log-determinant improvement over deployed SCAT32 exceeds the 10^{-2} bar by roughly two orders of magnitude (median 1.43, range 1.19–1.87 per dimension, found by the frozen support-expanding primal screen); the 181 unresolved cells returned an infeasible cutting-plane master under the frozen numerical protocol and are not certified — which is not the same as scientifically refuted. Under the frozen failure-branch rule no categorical near-optimality claim and no collapse claim is made. Descriptively, the counterexample cohort is confirmatory evidence that the balanced design's scene-adapted geometry headroom survives the full deployed conditioning stack — now over fresh confirmatory scenes and the complete stack, consistent with the development-only dose-only probe of Section 3.3. Per-cell statuses, gaps, residuals, cut counts, and wall times are in the disclosure ledger (supplement S5.2).

5.4 Context arms and design diagnostics

At the deployed operating point ($\bar{\rho} = 0.60$, $\nu = 2000$) the frozen context arms complete the occupancy ladder that motivated the $k = 32$ scattered support: mean radiometric PSNR is 17.12 dB for SCAT32-060, 14.18 dB for the scattered $k = 16$ SCAT16, and 9.63 dB for the clustered LBLOB16, placing the deployed design above both descriptive references at matched load (Table 6) and consistent with the frozen development-selection rule (Section 4). The dose-relaxed and dose-constrained continuous OED solutions, retained as no-endpoint design diagnostics, and the development-cell guard-versus-α collapse (ADAPTIVE_COLLAPSE_UNDER_GUARDS under the joint dose/A-risk/spectral guard stack, with the path-specific descriptive PATH_FEASIBLE_ALPHA diagnostic) feed the Act III figure (Figure 1); their full tables and the dose-only development probe fields are in supplement S5.3.

[M1 (Act III, five panels): (a) dose deviation, A-risk, and spectral guard versus path-specific α on a DEV representative, labeled development evidence; (b) the constructive dose-only primal lower bound (development-only: per-cell $\Delta \log\det V/r$ of the probe designs vs. deployed SCAT32); (c) confirmatory G_{stack}/r and terminal-status distributions over the full-stack class, labeled `FULL_STACK_CERT_PASS` (480 cells); (d) ridge-versus-0.60 fixed-dwell reconstructions and per-scene quality gains at $\nu = 2000$; (e) a schematic distinguishing the photon-budgeted and time-limited resource corners.]

Figure 1: Act III: information rewards scene-adapted geometry, uniform dose alone does not eliminate that headroom (development evidence, panels a–b), the conditioning stack defines the deployable class tested by the full-stack certificate (panel c), and global operating-point control survives orthogonally (panels d–e). Panels fill from the frozen M1 analyzer after tag `m1-freeze`.

Table 6: Cross-arm summary at the primary dwell $\nu = 2000$ (descriptive; means over 24 scenes $\times 5$ seeds). Achieved $\bar{\rho}$ and incident-dose ratio are read from the imaging shards; the incident-dose ratio is normalized to the SCAT32-060 comparator. Ceiling fraction is the predicted count-saturation probability $p_{\mathrm{ceil}} = \mathrm{P}(N = \nu)$ from the frozen renewal kernel at the achieved mean load (realized dark-count fraction is 0 for every arm). Mean J_{exact} is the exact per-slot count information evaluated at the achieved mean load via the same frozen kernel. The gates are the RIDGE-SCAT32 primary and speed comparisons and the deployed-design certificate; "—" marks a paired metric that does not apply to a reference or context arm.

Arm	Role	achieved $\bar{\rho}$	inc.-dose ratio	ceiling frac.	mean J_{exact}	$\Delta Q^{\mathrm{ridge}}$ / S_{gate}
SCAT32-SAFE	safe ref. / speed base	0.05	0.08×	$< 10^{-6}$	0.048	—
SCAT32-060	primary comparator	0.60	1.00×	$< 10^{-6}$	0.375	—
RIDGE-SCAT32	ridge policy (gated)	22.25	37.1×	$< 10^{-6}$	0.935	+1.87 / 0.276
SCAT16	context	0.60	1.00×	$< 10^{-6}$	0.375	—
LBLOB16	context	0.60	1.01×	$< 10^{-6}$	0.375	—

6 Discussion: the operating handbook

The campaign exposes two conjugate resource regimes of the same dead-time information map. Under a linear incident-photon budget the per-photon efficiency $J_{\mathrm{exact}}(\rho, \nu)/\rho$ falls monotonically, so a budget-constrained design rations photons onto lower loads and *avoids* the ridge; the ridge belongs to time-constrained operation, where power is available and the only scarcity is dwell. Reading the flux ridge and the design allocation as one program clarifies that the large fixed-dwell gains of the companion are an operating-point effect, not evidence that a design must spatially redistribute dose into the ridge.

The frozen verdicts are settled: the ridge operating-point primary *passes* (`RIDGE_OPERATING_PASS` = PASS, median 1.87 dB), the ridge speed secondary *fails* as a preregistered negative (`RIDGE_SPEED_PASS` = FAIL, median S_{gate}0.276), and the full-stack certificate *fails* to certify (`FULL_STACK_CERT_PASS` = FAIL, 0/480 cells certified); the campaign is preregistered to license only the governing ruling's frozen conclusions. Global operating-point control remains the deployed positive arm; geometry selection is now known to retain dose-feasible headroom outside the conditioning stack (Section 3.3), so operating-point control and geometry adaptation are not presented as an exhaustive either–or. Because `FULL_STACK_CERT_PASS` did not pass, the certificate licenses no categorical near-optimality statement and no collapse claim: across the 480 confirmatory cells, 0 certified, 299 counterexample, and 181 numerically unresolved (Section 5), so within the frozen full-stack class balanced SCAT32 is *not* certified near-optimal, and the confirmatory counterexample cohort descriptively shows that scene-adapted geometry headroom survives the full deployed conditioning stack — a localized, better-defined open problem, not a rescued outcome. The certificate is comparator- and subspace-relative: it is stated relative to the frozen expanded dictionary, the local pre-scan estimate, the declared resource model, and the directions SCAT32 resolves. The line's thesis, in the frozen wording of the governing ruling, is this. Uniform dose controls cumulative exposure but does not eliminate scene-adapted geometry selection. The operative restriction is the full deployed conditioning stack: balanced SCAT32 is tested for certified near-optimality within that frozen class, while time-limited image gains are tested separately by global ridge tracking.

The primary gain is scene-dependent, and its one negative-median family is diagnostic: on `contour` the

Table 7: Operating handbook for dead-time-limited single-pixel imaging. The engagement index Γ is descriptive (the companion's computable engagement criterion [1]), never a confirmatory gate.

Step	Rule
1. Characterize	measure τ and the hold-time c_v from time-tagged recovery intervals
2. Set the load	jitter-capped optimum from $c_v^2\rho^2(1+2\rho)=1$ (long window) or $(6\nu)^{1/3}(1+12\nu c_v^2)^{-1/3}$ (finite window); never operate a jittered detector at the deterministic ridge
3. Check engagement	common 52-pattern pre-scan $\rightarrow$ measured C_u; $\Gamma = C_u\sqrt{\nu\rho/(1+\rho)} \gtrsim 1$ marks the photon-limited regime
4. Pick patterns	photon-budgeted: the balanced SCAT32 design, tested for certified near-optimality within the full deployed conditioning class ($G_{\text{stack}}/r \le 10^{-2}$); time-budgeted: RIDGE-SCAT32 — one global multiplier to $\rho_R(\nu)$ (Section 3.7)
5. Doctrine	photon-constrained $\rightarrow$ stay below the ridge (J_{exact}/ρ falls); time-constrained $\rightarrow$ track the jitter-capped ridge

balanced 0.60 design outperforms the ridge on three of four instances (median $\Delta Q^{\text{ridge}} = -4.18$ dB, versus +6.37 on `maze`), the separating covariate being bucket-brightness contrast (losing instances $C_u \in [0.21, 0.25]$ against the winning instance's 0.33; across all 24 scenes ΔQ^{ridge} correlates with C_u at +0.64), with no guard, trust, dark-count, or ceiling flag firing, consistent with the alignment condition of Section 3.7 under which a single global flux multiplier is least effective where bucket brightness is weakly aligned with directional value; the residual family-geometry dependence — `contour` loses where an equally low-contrast `chirp` scene gains — is left as a hypothesis.

The jitter-capped ridge is the practical caution. The cube-root ridge is a property of the zero-jitter critical manifold; the moment a detector's hold time carries a coefficient of variation c_v, the long-window optimum is finite — the exact cubic $c_v^2\rho^2(1+2\rho)=1$ of Theorem 3, scaling as $2^{-1/3}c_v^{-2/3}$ — and driving past it is actively harmful: at the deterministic ridge ($\rho = 22.3$), 5% jitter already forfeits about 55% of the scalar count information in the refined sweep (Table 1). A deployment must therefore locate its operating point against the jitter-limited ceiling of the real detector, not the deterministic ridge; the safe conclusion is conditional on measured timing jitter.

Together, the law and the design principle compress into the operating handbook of Table 7: characterize the detector, let the law set the load ceiling, check engagement with the pre-scan, and pick the pattern recipe by which resource is scarce.

Two scope boundaries stand. All results are simulation-based, and every certificate is stated relative to the frozen dictionary, the local pre-scan estimate, and the declared resource model; an optical extension along the companion's discussion — a coarse complex-field estimate synthesizing an optical displacement that removes common-mode photons before dead-time loss — remains a natural bench direction [1].

The law itself, however, is bench-testable now, with no imaging required. For a real single-photon counting module, the hold-time coefficient of variation is directly measurable from time-tagged recovery intervals; the theory then predicts a count-information peak at the root of $c_v^2\rho^2(1+2\rho)=1$ rather than at the deterministic ridge — for $c_v = 0.05$, a peak near $\rho \approx 5.7$ instead of ≈ 22 at $\nu = 2000$, with about half the information forfeited if the deterministic ridge is used instead (Table 1). Sweeping source power at fixed dwell and estimating the count information from repeated frames traces $J(\rho)$ directly; this single-detector power sweep is the immediate bench experiment for the line.

7 Reproducibility

The design formulation, guards, and campaign follow six frozen rulings: the dead-time OED formulation and equivalence certificate (R10), the ridge-zone load architecture and incident-budget conjugacy (R11), the M1 freeze audit (R13), the unification ruling (R14, issue "R14 ruling: unification architecture"), whose theorem hierarchy Section 2 follows, the safety-constrained refreeze (R17, issue "R17 ruling: safety-constrained OED collapse and coherent refreeze"), which retired the adaptive imaging arm and re-architected the campaign around the ridge operating-point primary and a structural near-optimality certificate, and the full-stack localization (R18, issue "R18 ruling: full-stack certificate localization and final refreeze"), which redefined the confirmatory certificate over the full deployed conditioning stack (`FULL_STACK_CERT_PASS`), admitted the dose-only constructive probe as development-only evidence, and froze the corrected claim discipline of Section 3. Exact-law peak predictions for the refined jitter sweep, including the two out-of-sample rows, were registered before the sweep completed (`docs/R14_PREREGISTERED_PREDICTIONS.md`, commit `8a627bb`) and confirmed in three stages (grid verdict `1d8d7aa`; local zoom `4d8310d`; pooled Colab replication `bd731d5`), with the intermediate $k = 4/3$ hypothesis preregistered at `5e51088` and refuted at `1a32758`. The campaign freezes at tag `m1-freeze` once the R13 checklist passes outcome-blind self-tests; the companion is frozen at tag `paper1-freeze-v1`. Source-of-record documents for every frozen constant are cited by ruling number in the block comments of the manuscript source; full machinery and proofs are in the supplement (`supplement_m1.tex`). All code, frozen preregistration documents, per-cell manifests, and evidence bundles are available at [repo URL wording — user decision]; funding and acknowledgments are [funding/acknowledgments — user decision].

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